Rethinking Artificial Intelligence creativity and ideation systems

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Abstract

The visual and media arts community faces a significant challenge from the rise of generative visual AI, which poses a threat to their livelihoods. The current prompt-based generative AI (genAI) systems, such as Stable Diffusion, Midjourney, and DallE, are limited in their ability to provide sustainable financial opportunities for artists and designers, while also producing output that homogenizes unique cultural perspectives and styles. The importance of this topic lies in the cultural value that artists and designers bring to society, serving as a medium for expressing and preserving diverse traditions, beliefs, and values. We discuss a new approach to genAI and revised technological direction, as the current trajectory of AI development has the potential to displace human creators and devalue their skills.

1 Introduction

We propose techniques and have begun implementing a prototype system to alter the current direction of genAI development and to create a pathway for creative professionals to utilize these systems to their advantage, both creatively and monetarily. The goal is to enhance (rather than constrain) the creative process of artists and designers by allowing them to ideate their own creative visions into a final artistic output of personal and/or societal significance and in the creative's own unique style. This toolkit also includes sustainable pathways for artists and designers to sell and/or license their work.

2 Rethinking Gen AI for more artist-centric creativity and ideation

We have a multi-year research in state-of-the-art in cognitive-based Deep AI techniques for creativity, and suggest these be applied to the creation of more artist-centric AI tools. This research goes beyond current efforts by utilizing our research in diffusion transformer and Large Language Modeling (LLM) systems, to create a system for artist-centric creative processes and workflows. Including:

1. Diffusion transformer based systems (Peebles, Xie 2023) and more aesthetic meaning centric datasets – where we use our cognitive modelling of creativity (DiPaola, McCaig 2016; Utz, DiPaola 2020), aesthetic emotions (Abukhodair, … DiPaola 2023; Yalcin, Abukhodair, DiPaola 2014, 2019) and bringing more human-centric meaning into datasets (and captions) (Salevati, DiPaola 2015a, 2015b, 2016, 2017)
2. Innovative new forms of LLMs and their fine tuning (He et al. 2023) to bring that deep meaning on some of these same areas: aesthetics, style, and emotion. Our new LLM system (Gonzalez, DiPaola 2024) which we will continue improve, uses graph networks and new forms data augmentation systems rather than zero-shot generators. It uses robust cogni-

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tive and memory frameworks with data enrichment, episodic memory, and self-reflection (Gonzalez, DiPaola 2024).

3. Better forms of local training of models such as LORA and fine tuning (He et al 2023; Lin et al 2024; Choi, DiPaola, 2023ab) that can better encode an artistic style allowing new forms monetization and collaboration of an artist's body of work – where style and meaning are better built into the technologies.

We are now working with a prestigious design firm, FarmBoy Fine Arts (Farmboy) which ideates ideas for major hotels, hospitals and major car companies, as well as students from a well know art and design university, who with us are testing how they can explore new shared work of combined designer/artists through their style. While our prototype tools are well underway with this evaluation work, we believe there are many unanswered questions in this very new way of creative ideation by cognitively journeying through a n-dimensional creative latent space, so are documenting questions and issues in this paper. Throughout this stage of the project, researchers need to work out the following aspects of this new creative system:

1. What are the best new AI techniques and parameters for this type of LoRA (local training of a subset of creative work) training in style?
2. What image captioning techniques work best – captioning at training time or what is prompted at creation time? Do you use keywords or full design briefs to caption?
3. What size of dataset (input artworks) works best to maintain the style of an artist with a balance between style and variation for the designer to create?
4. How do these LoRAs work for the artist – can an artists create via their own personal dataset (or varied datasets/models) of their own work? How does this benefit individual artists?
5. How exactly does financial compensation work when a company contracts to use a LoRA of a specific affiliated artist/designer? (Farmboy already has years of experience with revenue sharing with their affiliated artists with several existing plans that will be tested and modified as needed)

Current barriers impeding a solution include the limitations of prompt-based generative AI systems, which homogenize unique perspectives and styles, and the lack of sustainable financial opportunities for artists and designers within these systems. Current genAI systems like Midjourney and DalleE are based on using a text prompt that a user types into the system to generate an image or video, prose. These prompts are not suited for many creative forms of expressions, as they are both limited as well as based on the original captions of the input images (or stories) in the dataset. These captions or keywords can be very superficial. Moreover, these text prompt systems do not allow artists to easily journey through a wide n-dimensional aesthetic space (the latent space of the dataset), which is necessary for creative exploration to achieve the result they envision in their mind. In current systems, when the user changes the prompt due to dissatisfaction with current output – the output "jumps" around to another part of the search space (aesthetic latent space), preventing a slow and intentional exploration.

Our recommendations and prototype system allows for deeper meaning attributions that are linked to emotions, aesthetics, human philosophy, cultural history, and cognitive systems of art and vision (color palette, style, shapes, abstraction).

Our existing prototype allows an artist to move from a current idea iteratively from image to image towards their final goal. Furthermore, it allows them to label their creative travel, save the data and map out the journey, allowing the user to move forward or backwards through their creative journey. Preliminary work has shown that the labeled 3d mapping helps artists to better visualize their current creative journey. Our novel strategy involves developing an AI-powered toolkit that enhances the creative process, allows for open creative journeying, and enables artists to license their work for revenue-sharing.

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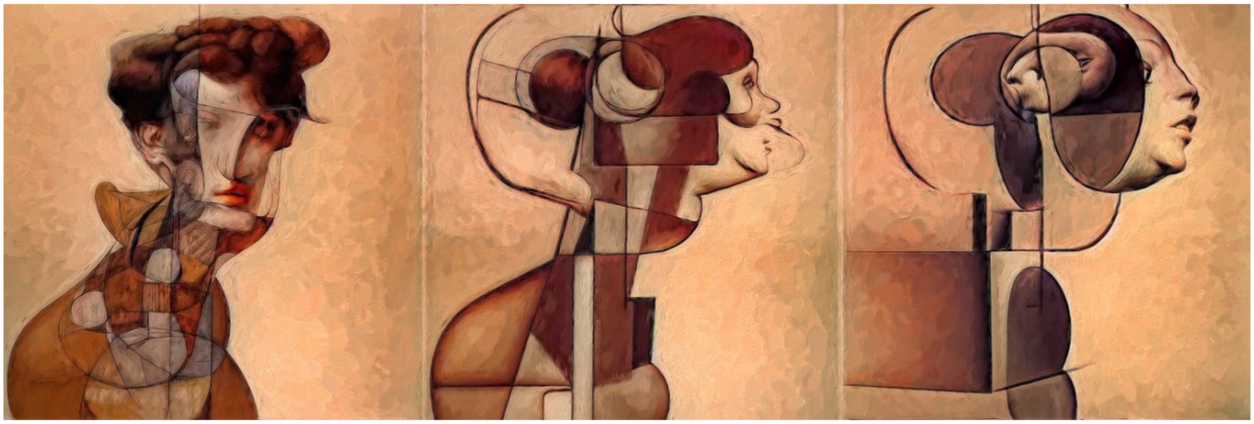
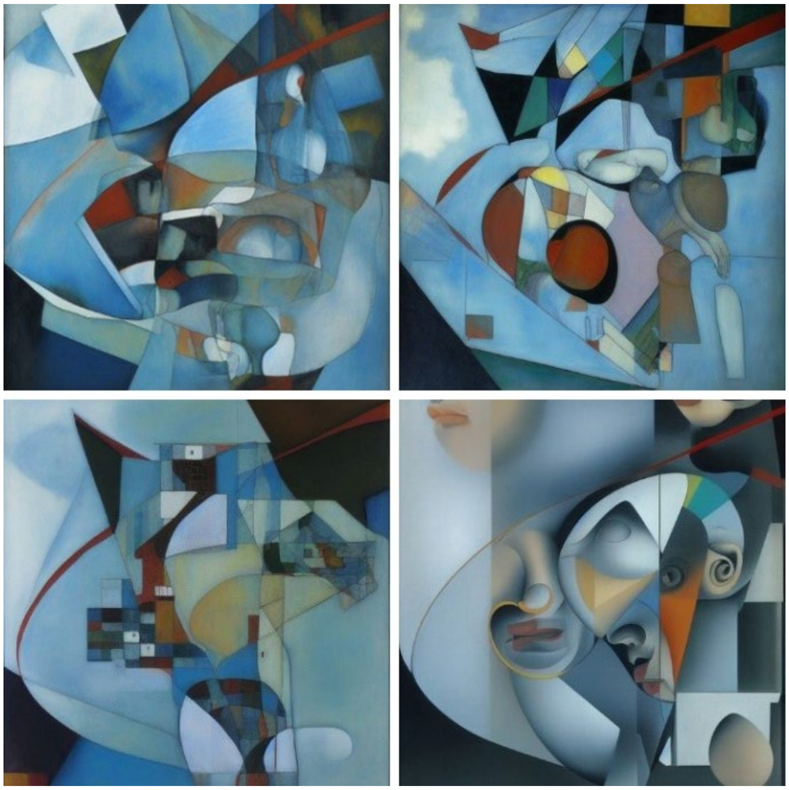
Supplemental material DiPaola

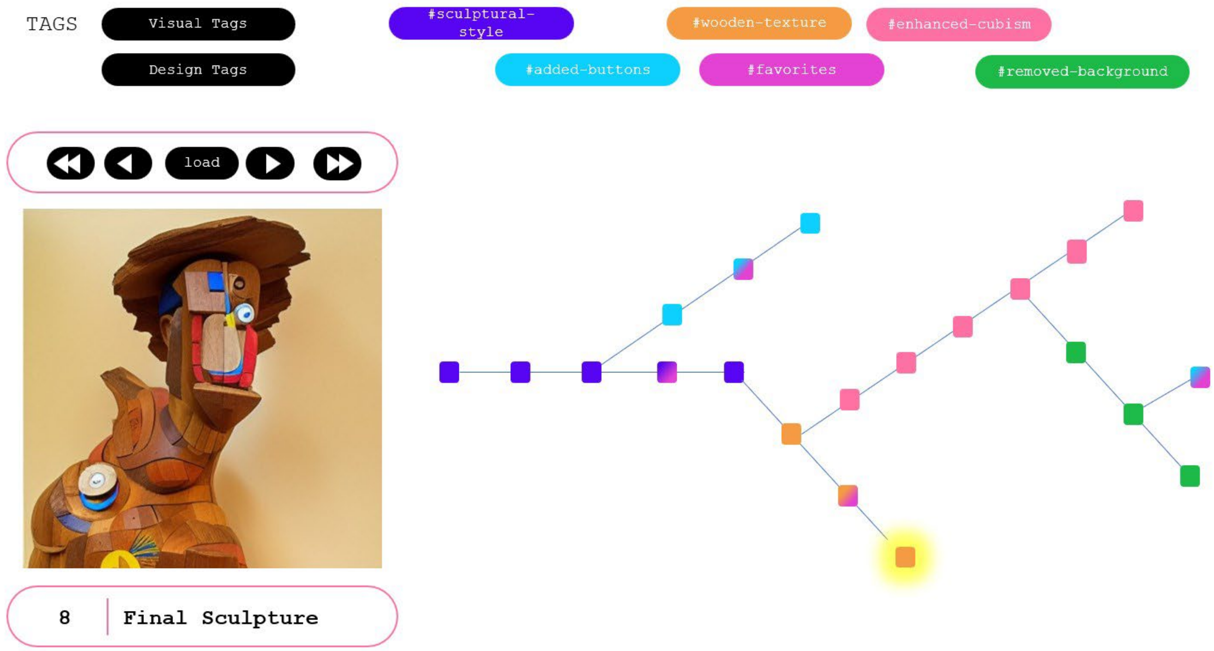


An examples demonstrating movement through concept or style using our tool allowing for experimentation through complex associations along any axis in image-concept in latent space.



A grouping of images from the same region of latent space. The intersection of several aesthetic vectors reveals a diverse region of related affective stimuli that is not necessarily transparent to the interacting artist.

With our system using a LORA within our system, with a certain style - using the process to move through metaphor space – where “virtual cutout paper feather strokes” compose together to become, first birds, than trees and resolves by journey back in time and space over 100s of images to morph of a “bird tree fly” hybrid landscape



Our system can zoom in , and pan around latent space – creating a archived and live creative journey. It then creates a dynamic map of every part of the journey. The mapping system shows forks that are taken (with tags and coloring) with in your creative decisions and allowing you to go back and forth – which allows you to better be free to experiment with ideation. Here you can click on any part of the map to bring up the user label (ie Final Sculpture) and output from any part of your paths as you journey.